信 息 学 院

《社交网络技术与应用》

作业一 多模态模型部署

实验题目 多模态模型部署

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1 实验介绍

在当今的人工智能浪潮中，多模态大模型已经成为连接视觉与自然语言的核心技术。本节实验我们将跳出纯文本的局限，使用 *Hugging Face* 官方开源的新一代极轻量级多模态模型：*SmolVLM-500M-Instruct*。

实验目的

- 掌握 Python 虚拟环境的创建与依赖隔离原则。
- 熟悉当前工业界主流的大模型生态库 transformers 的最新 API 规范。
- 解决不同硬件架构下的深度学习计算环境配置问题。
- 成功在本地跑通“图片输入 - 模型理解 - 文本生成”的 AI 推理流水线。

实验清单

实验	简述	难度	软件环境	开发环境
基础任务：SmolVLM部署	跑通 500M 轻量级多模态大模型的本地部署，实现基础的“单图+文本”分析。	初级	Python 3.10	本地PC
进阶任务：InternVL 3.0 部署	部署目前前沿的 InternVL 3.0 架构，体验动态分辨率预处理与多轮流式对话。	中高级	Python 3.10	本地PC

2 实验环境验证（20%）

目的： 验证虚拟环境配置正确，且成功识别计算设备。

终端截图： 请提供一张包含终端命令行的截图。要求截图中必须显示激活的（lab_你的学号）环境前缀 ，并在终端中执行以下 *Python* 代码的结果：

```
import torch

print(torch.cuda.is_available())

print(torch.__version__)
```

```
(E:\Social Media Technoledge and application\lab_3780) E:\Social Media Technoledge and application>python
Python 3.10.19 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 16:41:31) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> import torch
>>> print(torch.cuda.is_available())
True
>>> print(torch.__version__
... )
2.12.0.dev20260305+cu128
>>>
```

3 SmolVLM-500M 部署与测试（60%）

目的： 验证模型成功下载并在本地运行推理流水线 。

官方示例测试：运行手册中提供的代码，测试自由女神像图片。请提供控制台输出 *AI* 回答：部分的完整截图 。

```
(E:\Social Media Technoledge and application\lab_3780) C:\Users\koshitann>python e:/code/vscode/Python/SocialMedia/run_smolvlm.py
正在使用设备: cuda
正在下载测试图片...
正在加载模型权重...
processor_config.json: 100%|          | 68.0/68.0 [00:00<00:00, 68.1kB/s]
E:\Social Media Technoledge and application\lab_3780\lib\site-packages\huggingface_hub\file_download.py:129: UserWarning: `huggingface_hub` cache-system uses symlinks by default t
o efficiently store duplicated files but your machine does not support them in C:\Users\koshitann\.cache\huggingface\hub\models--HuggingFaceTB--SmolVLM-500M-Instruct. Caching file
s will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment v
ariable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://doc
s.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
  warnings.warn(message)
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
chat_template.json: 100%|          | 429/429 [00:00<?, ?B/s]
preprocessor_config.json: 100%|          | 486/486 [00:00<00:00, 484kB/s]
config.json: 7.34kB [00:00, 7.37MB/s]
tokenizer_config.json: 28.2kB [00:00, 28.2MB/s]
vocab.json: 801kB [00:00, 2.02MB/s]
merges.txt: 466kB [00:00, 5.65MB/s]
added_tokens.json: 4.74kB [00:00, 4.75MB/s]
special_tokens_map.json: 1.07kB [00:00, 1.07MB/s]
tokenizer.json: 3.55MB [00:00, 32.9MB/s]
`torch_dtype` is deprecated! Use `dtype` instead!
model.safetensors: 100%|          | 1.02G/1.02G [06:18<00:00, 2.68MB/s]
Loading weights: 100%|          | 489/489 [00:00<00:00, 2183.35it/s]
generation_config.json: 100%|          | 136/136 [00:00<00:00, 89.6kB/s]
模型思考中...

==================================================
The image depicts a cityscape featuring a prominent landmark, the Statue of Liberty, prominently positioned on Liberty Island. The statue is a green, humanoid figure with a crown
atop its head and is situated on a small island surrounded by water. The statue is characterized by its detailed sculpting, with its left arm raised and its right arm resting on t
he island. The statue's face is serene, with a calm expression, and it is surrounded by a series of buildings, which are typical of urban settings.

In the background, the cityscape is filled with numerous skyscrapers, indicating a modern urban environment. The buildings vary in height and design, with some being taller and mo
re modern, while others are older and more traditional. The sky above is clear, suggesting good weather conditions, and the sun is shining, casting a warm glow over the scene.

The water surrounding the island is calm, with small waves visible, indicating a relatively calm and peaceful environment. There are a few boats and small ships visible in the wat
er, adding to the sense of a bustling city.

The image captures the essence of a city that is both vibrant and orderly, with the statue serving as a focal point and the surrounding buildings providing a sense of scale and de
pth. The overall composition of the image is balanced, with the statue and the city in the foreground, while the background is dominated by the modern urban landscape.

### Analysis:
1. **Statue**: The Statue of Liberty is a well-known symbol of freedom and democracy, and its presence in the cityscape signifies the city's commitment [illegible] values.
2. **Cityscape**: The modern cityscape with skyscrapers and a calm water body suggests a developed urban area, likely in a major metropolitan region.
3. **Weather**: The clear sky and warm sunlight indicate favorable weather conditions, which are essential for city life.
4. **Time of Day**: The image captures the sun at a low angle, suggesting either early morning or late afternoon, which is a common time for city lights [illegible]e.

### Conclusion:
The image effectively captures the essence of a modern city, with the Statue of Liberty serving as a symbol of freedom and democracy, and the surround[illegible]scape providing
 a sense of scale and depth. The calm water and clear sky indicate a pleasant weather condition, which is essential for city life. The overall compositi[illegible]image is balanced
, with the statue and the city in the foreground
==================================================
```

自定义医学影像测试：

操作： 请在网上寻找一张医学影像图（如 *X* 光片、*B* 超、*CT* 图像等）或任意一张包含复杂信息的图片，替换掉代码中的测试图片。

提问： 将代码中的 *user_question* 修改为你自己设计的英文问题。

结果提交： 请将你使用的原图以及控制台完整的问答输出截图贴在下方。

```
==================================================
The image is a medical scan, specifically an MRI (Magnetic Resonance Imaging) scan of the chest. The scan is in blue and white, which is typical for such imaging techniques. The s
can shows the internal structures of the lungs, including the bronchi (airways) and the heart.

### Detailed Description:
1. **Lungs**: The scan shows the lungs in their entirety, with the bronchi and heart visible. The bronchi are the main airways that lead to the lungs, and the heart is located wit
hin the thoracic cavity, which is the space inside the chest.

2. **Bronchi**: The bronchi are visible as thin, white lines branching out from the heart. The bronchi are the main airways that lead to the lungs.

3. **Heart**: The heart is visible as a dark area in the center of the scan. The heart is a muscular organ that pumps blood throughout the body.

4. **Lungs**: The lungs are visible as a light blue area in the center of the scan. The lungs are the two main organs in the chest cavity, responsible for gas exchange and oxygena
tion of blood.

5. **Soft Tissue**: The scan shows the soft tissue surrounding the lungs, including the diaphragm (a muscle that separates the chest cavity from the abdominal cavity).

6. **Airway**: The scan shows the airways, including the trachea (a tube that connects the bronchi to the lungs), bronchioles, and the alveoli (tiny air sacs where gas exchange oc
curs).

7. **Blood Vessels**: The scan shows the blood vessels, including the aorta (the largest blood vessel in the body) and the pulmonary arteries (which carry blood to the lungs).

### Analysis:
This scan provides a detailed view of the lungs and heart, which can be useful for diagnosing various conditions, such as pneumonia, lung cancer, or heart disease. The presence of
 fluid in the lungs, known as pleural effusion, can indicate conditions like pneumonia or pleural effusion. The presence of fluid in the heart can indicate conditions like heart f
ailure.

### Conclusion:
The image is a medical scan of the chest, providing a detailed view of the lungs and heart. The scan is crucial for diagnosing and monitoring various conditions, including pneumon
ia, heart disease, and lung cancer. The presence of fluid in the lungs can indicate conditions like pleural effusion, which can be treated with
==================================================
```

问题：Clearly describe the image. What the image shows? Where it would be found and used for?

模型能力评估： 结合你的测试结果，简要评价这个 5 亿参数（500M）的极小模型在你提供的专业图片上的表现。它是否准确识别了图像主体？是否存在“幻觉”（胡编乱造）？

该模型准确识别了图像主体，判断出这是一张胸部医学影像，并指出了肺部结构、支气管等，用于肺部和心脏等部位相关疾病的诊断也基本正确。

存在“幻觉”，如将图像识别为 MRI 扫描，实际上本图是 CT 扫描；将图像中央识别成了心脏；肺部积液等病理现象疑似无中生有。

该模型仅能对图像主题进行大致的识别，在细节上的识别效果差，不适用于医疗等专业领域。

3 工程反思与代码剖析（20%）

目的： 培养独立排查工程环境报错的能力，以及对深度学习底层参数的理解。

排错记录： 在本次实验的网络配置、依赖库安装和代码运行过程中，你遇到了哪些报错？你是如何解决的？（如果没有遇到报错，请写“无”，并简述你认为如果是纯 CPU 电脑，在这套流程中最容易卡在哪一步）。

无。

最容易发生在加载模型权重阶段，GPU 有专门的 CUDA 核处理张量，而纯 CPU 仅依赖通用计算核心，对于模型权重加载的计算量有限，尤其是在电脑同时运行有多个进程时，可用内存被进一步压缩，计算速度进一步减慢。

参数理解： 在初始化模型的代码中，有这样两行关键的硬件适配参数：

```
torch_dtype=torch.bfloat16 if DEVICE == "cuda" else torch.float32,
_attn_implementation="sdpa" if DEVICE == "cuda" else "eager",
```

请查阅相关资料，分别简述 bfloat16（半精度浮点数）和 sdpa（Scaled Dot-Product Attention）在工程部署上的作用是什么？它们是如何帮助我们在普通的个人电脑上更顺畅地跑起大模型的？

bfloat16 是一种专为 AI 计算设计的半精度浮点数格式，占用 2 字节，相比于标准 FP32 单精度占用 4 字节，在内存层面将模型权重、计算过程中的张量数据体积直接减半，大幅降低内存占用，在计算层面显著减少张量运算耗时。在精度上保留了 FP32 相同的指数范围，仅缩减尾数位精度，既避免 FP16 容易出现的数值溢出问题，又能在 AI 任务中保证精度损失可接受。为普通的个人电脑降低了模型权重占用的内存，提升计算速度。

SDPA 是 Transformer 模型核心模块的优化版实现，对比 eager 模式，在算法层面上内置缩放、mask 处理、数值稳定性优化，避免手动实现注意力时的冗余计算。在硬件层面深度适配 CUDA/CPU 指令集，将注意力计算的多个步骤整合成“原子化操作”，减少内存读写次数。注意力层是大模型推理中最耗时的环节，在 GPU 场景下，SDPA 调用 CUDA 核心的专用注意力计算接口，避免 eager 模式下的 CPU-GPU 数据来回拷贝，让注意力计算速度翻倍。在 CPU 场景下则使用 eager 模式。